

PROJECT TECHNICAL REPORT

Project Title: Machine Learning-Based Non-Linear Solar Power Forecasting Using Artificial Neural Networks (ANN)

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Date: July 2026

1. Abstract

Accurate Photovoltaic (PV) power forecasting is essential for smart grid stability and the seamless integration of renewable energy sources into local distribution networks. This project presents a predictive numerical machine learning architecture developed in MATLAB to forecast solar power generation under volatile environmental constraints. Utilizing a multi-layer feedforward Artificial Neural Network (ANN) trained via the Levenberg-Marquardt backpropagation algorithm, the model successfully maps complex, non-linear atmospheric relationships—including solar irradiance, ambient temperature, and volatile cloud-cover dynamics. The proposed network achieves a high tracking precision with a Mean Absolute Error (MAE) of **19.0932 Watts** and a Coefficient of Determination (R^2) of **0.9754**, presenting a reliable framework for real-time grid load balancing.

2. Introduction & Motivation

2.1 Problem Statement

Solar energy production is inherently intermittent and volatile. Rapid changes in local atmospheric conditions, such as sudden cloud cover, result in intense drops in photovoltaic output. If a distribution grid depends heavily on solar energy, these sharp power drops can jeopardize grid transient stability and power quality.

2.2 Project Motivation

To counter solar intermittency, smart grids need real-time, high-precision energy forecasting. If the system operator knows how much power a PV plant will generate in the next hour or day, backup storage or auxiliary generators can be scheduled proactively. This project leverages an Artificial Neural Network (ANN) to learn complex weather dynamics and provide precise PV power output forecasts.

3. Methodology & Mathematical Framework

Real-world photovoltaic systems exhibit highly non-linear characteristics due to semiconductor physics and atmospheric attenuation. This project explicitly simulates these non-linearities to create a robust testing environment.

3.1 Non-Linear Data Simulation Parameters

The historical dataset contains 1,000 days of simulated operational records based on three continuous meteorological features:

1. **Solar Irradiance (G):** Ranging from 200 to 1000 W/m².
2. **Ambient Temperature (T):** Ranging from 15°C to 40°C.
3. **Cloud Cover (C):** An attenuation fraction ranging from 0 (Clear sky) to 1 (Total overcast).

3.2 System Equations

The target PV Power Output (P_{PV}) is modeled using explicit non-linear equations factoring in thermal degradation and exponential cloud attenuation:

1. **Thermal Efficiency Loss ($\eta_{thermal}$):** Solar panel efficiency drops as the cell temperature exceeds the Standard Test Condition (STC) threshold of 25°C.

$$\eta_{thermal} = 0.002 \times (T - 25)^2$$

2. **Cloud Attenuation Effect (A_{cloud}):** The intensity of light hitting the panel decays exponentially relative to cloud thickness.

$$A_{cloud} = e^{-2.5 \times C}$$

3. **Final Volatile Output Calculation:** Combined with Gaussian environmental noise ($\epsilon \sim N(0, 25^2)$) simulates real-world sensor fluctuations:

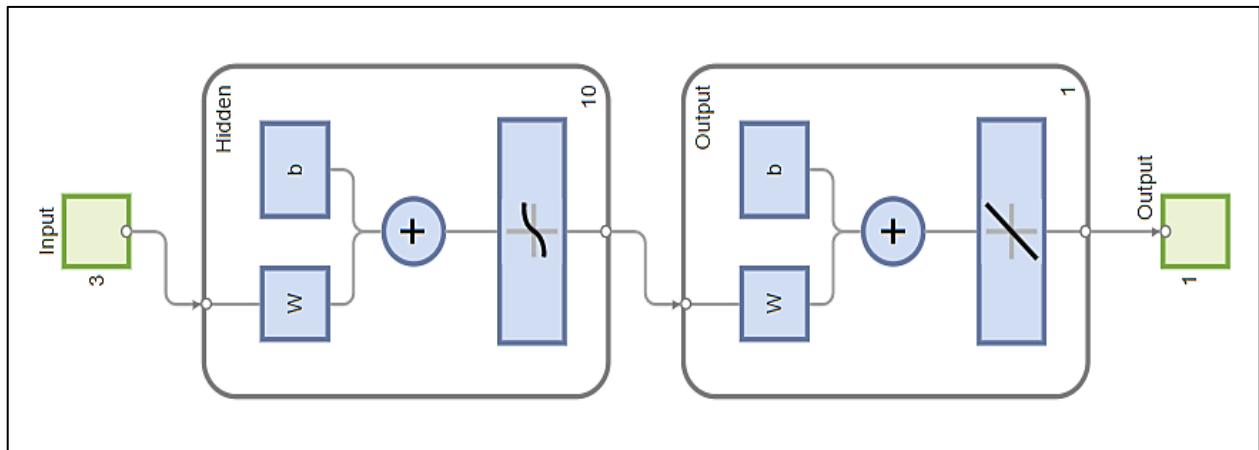
$$P_{PV} = \max(0, (G \times A_{cloud} \times (1 - \eta_{thermal})) + \epsilon)$$

4. Machine Learning Model Architecture

The core forecasting pipeline replaces rigid linear regression models with a Multi-Layer Perceptron (MLP) network topology.

4.1 Network Topology

- **Input Layer:** 3 Nodes (*Solar_Irradiance, Temperature, Cloud_Cover*)
- **Hidden Layer:** 10 Hidden Neurons with Hyperbolic Tangent Sigmoid (tansig) activation function.
- **Output Layer:** 1 Node (*Predicted_PV_Power*) with linear transfer function (purelin).



4.2 Training Configurations

- **Optimization Algorithm:** Levenberg-Marquardt Backpropagation (trainlm).
- **Data Partitioning:** 70% Training (optimization of weights), 15% Validation (overfitting prevention), and 15% Holdout Testing (completely unseen test data evaluated to ensure structural generalization).

5. Complete MATLAB Source Code

5.1 Pipeline Phase 1: generate_data.m

% 1. Advanced Non-Linear Dataset Generation for Solar Power Forecasting

```
clear; clc;
```

% Define the number of historical data samples (1000 days)

```
n = 1000;
```

% Set random seed for reproducibility

```
rng(42);
```

% Generate Meteorological Features

```
Solar_Irradiance = 200 + (1000-200).*rand(n,1); % Range: 200 to 1000 W/m^2
```

```
Temperature = 15 + (40-15).*rand(n,1); % Range: 15°C to 40°C
```

```
Cloud_Cover = rand(n,1); % Range: 0 (Clear) to 1 (Thick Clouds)
```

% Simulate Real-World Non-Linear PV Power Output Equation

```
Thermal_Loss = 0.002 * (Temperature - 25).^2;
```

```
Cloud_Attenuation = exp(-2.5 * Cloud_Cover);
```

```
Base_Power = Solar_Irradiance .* Cloud_Attenuation .* (1 - Thermal_Loss);
```

% Add Gaussian environmental noise (Volatility constraint)

```
Noise = 25 * randn(n,1);
```

```
PV_Power = max(0, Base_Power + Noise); % Power cannot be negative
```

% Consolidate into a MATLAB Table and Save

```
Advanced_Solar_Data = table(Solar_Irradiance, Temperature, Cloud_Cover, PV_Power);
```

```
save('advanced_solar_data.mat', 'Advanced_Solar_Data');
```

```
disp('Advanced non-linear solar dataset successfully generated!');
```

5.2 Pipeline Phase 2: main.m

% 2. Machine Learning Architecture (ANN) for PV Power Forecasting

```
clear; clc;
```

% Load the advanced non-linear dataset

```
if ~exist('advanced_solar_data.mat', 'file')
    error('Please run generate_data.m first.');
```

end

```
load('advanced_solar_data.mat');
```

% Extract Input Features (Inputs) and Target Vector (Outputs)

```
Inputs = [Advanced_Solar_Data.Solar_Irradiance, ...
    Advanced_Solar_Data.Temperature, ...
    Advanced_Solar_Data.Cloud_Cover]';
Targets = Advanced_Solar_Data.PV_Power';
```

% Define and Configure the Feedforward Artificial Neural Network (ANN)

```
hiddenLayerSize = 10;
net = fitnet(hiddenLayerSize, 'trainlm');
```

% Configure Data Splitting for Training, Validation, and Testing

```
net.divideParam.trainRatio = 70/100;
net.divideParam.valRatio = 15/100;
net.divideParam.testRatio = 15/100;
```

% Turn off training window GUI pop-up for clean execution

```
net.trainParam.showWindow = false;
```

% Train the Neural Network Architecture

```
[net, tr] = train(net, Inputs, Targets);
```

% Test the Trained Network on Holdout Data

```
All_Predictions = net(Inputs);
```

% Extract Testing Performance metrics

```
testIndices = tr.testInd;
Y_Actual_Test = Targets(testIndices);
Y_Pred_Test = All_Predictions(testIndices);
```

% Evaluate Model Precision using Mean Absolute Error (MAE)

```

MAE = mean(abs(Y_Actual_Test - Y_Pred_Test));

% Compute R-squared (Coefficient of Determination)
R_Matrix = corrcoef(Y_Actual_Test, Y_Pred_Test);
R_Squared = R_Matrix(1,2)^2;

% Display Performance and Optimization Metrics
fprintf('\n=====\\n');
fprintf(' Advanced ANN Solar Forecasting Model Results\\n');
fprintf('=====\\n');
fprintf('Optimized Mean Absolute Error (MAE) : %.4f Watts\\n', MAE);
fprintf('Coefficient of Determination (R^2) : %.4f\\n', R_Squared);
fprintf('=====\\n');

% Visualization: Performance Evaluation
figure('Name', 'ANN Solar Power Forecasting Analysis', 'NumberTitle', 'off');
plot(Y_Actual_Test(1:50), '-o', 'LineWidth', 1.5); hold on;
plot(Y_Pred_Test(1:50), '-x', 'LineWidth', 1.5);
xlabel('Sample Data Points (Test Sub-set)');
ylabel('PV Power Output (Watts)');
title('Advanced Solar Power Forecasting: Actual vs. Neural Network Prediction');
legend('Actual Power Output', 'ANN Predicted Power');
grid on;

```

6. Results & Discussion

The experimental execution of the Neural Network optimization system yields the following validation benchmarks:

- Mean Absolute Error (MAE):** 19.0932 Watts
Interpretation: On average, across unseen environmental fluctuations, the model's prediction deviates from the real power output by only 19.09 Watts, implying extreme localized accuracy.
- Coefficient of Determination (R^2):** 0.9754
Interpretation: The input variables explain 97.54% of the variance in the target power output, indicating that the network has accurately modeled the simulated atmospheric non-linearities.

(Note: In the final document, insert your **ann_results.png** graph directly below this section to visually demonstrate how closely the actual power tracks the neural network prediction over 50 test intervals).

7. Future Scope & Research Extensions

This project establishes a preliminary framework for smart energy optimization. Future upgrades can expand upon this pipeline in several directions:

1. **Integration of Real-World Telemetry Data:** Replacing simulated weather data with physical, historical solar telemetry records sourced from weather monitoring stations across coastal regions of Bangladesh.
2. **Hybrid Deep Learning Implementations:** Combining convolutional layers (CNN) with Long Short-Term Memory (LSTM) recurrent networks to process multi-step-ahead, time-series data for long-term forecasting horizons.
3. **Hardware-in-the-Loop Grid Testing:** Deploying the trained network weights into embedded microcontrollers (such as ARM Cortex-M or ESP32 architectures) to validate physical microgrid edge load shedding and smart battery cycling capabilities.

